Construction Management Software

Updated Product Scope & Technology Stack

A platform for projects, site operations, materials, labour attendance, progress evidence, receipts, and dealer requirements.

Product scope document

1. Project Overview

The proposed applicationwill centralize construction operations into one secure platform. Administrators will have visibility across projects, while site engineers will record daily site activity from a phone or tablet. The application will support multiple projects per engineer and maintain a traceable history of materials, labour, progress evidence, receipts, and purchase requirements.

Core objectives

- Replace disconnected spreadsheets, calls, and messaging threads with structured project records.
- Provide visibility into site progress, materials, labour, and pending requirements.
- Make field reporting simple for users with limited time or limited language education.
- Maintain evidence through photographs, audio notes, receipts, timestamps, and user identity.
- Generate reports and dealer-ready order-requirement PDFs.

2. User Roles and Access

Role	Main responsibilities	Access
Admin	Manage users, projects, engineers, materials, receipts, reports, and order requirements.	All projects and modules
Site Engineer	Manage assigned projects, submit progress, record materials, mark attendance, and upload evidence.	One or more assigned projects

Access principles

- One site engineer can manage more than one project from a single account.
- Engineers can access only their assigned projects.
- Admins can view and manage all project records.
- Approval and editing permissions can be separated by role.
- Important actions should be recorded in an audit trail.

3. Site Engineer App

Thesite engineer application willbe mobile-first and designed around daily site operations. The engineer selects a project, records the required information, attaches evidence, and submits the entry for administrative review.

Feature	Description
Multiple project management	A single engineer can switch between multiple assigned projects and sites.
Material income	Record materials received, item, quantity, unit, supplier, date, invoice/challan, and receipt.
Material outgoing	Record materials issued or consumed, quantity, destination, task, and remarks.
Daily labour attendance	Mark workers present or absent, working hours, labour category, subcontractor, and project allocation.
Labour notes	Add simple written notes about a labourer’s work, issue, requirement, or performance.
Audio recording	Record voice notes for labourers or site observations.
Daily progress photos	Capture and upload photographs linked to project, site, date, task, and daily report.
Daily progress report	Submit completed work, quantities, remarks, labour count, material movement, and photographs.
Receipt upload	Upload receipts, invoices, challans, and other material documents.

4. Admin App

Theadmin applicationwill provide a centralized view of all projects and operational records. It will also convert site requirements into organized purchase documents that can be sent to dealers.

Feature	Description
Project dashboard	View active projects, assigned engineers, recent progress, pending approvals, and requirements.
Material management	View material income, outgoing, current balance, transaction history, and reports.
Receipt visibility	Receipts uploaded by engineers automatically appear in the admin material section against the relevant project and transaction.
Material reports	Generate daily, monthly, and yearly reports for received, issued, consumed, returned, and balance quantities.
Labour reports	Review daily attendance, labour allocation, working hours, and manpower summaries.
Progress gallery	View daily progress photos by project, site, date, task, and engineer.
Notes and audio records	Access engineer notes and audio recordings linked to labourers or site activities.
Order requirements	Create requirements with item, quantity, unit, required date, project, and dealer.
Dealer order PDF	Generate a professional PDF with project details, required items, delivery information, and sign-off fields.

5. Key Workflows

Material workflow

Material requirement → Review → Material received → Receipt uploaded → Stock updated → Material issued/consumed → Report updated.

Daily site workflow

Engineer selects project → Records work → Marks labour attendance → Records material income/outgoing → Adds notes or audio → Captures progress photos → Submits daily report → Admin reviews.

Dealer requirement workflow

Admin selects project → Creates requirement → Adds items and quantities → Selects dealer → Generates order-requirement PDF → Downloads or sends PDF → Tracks status.

Suggested status lifecycle

Record type	Status lifecycle
Daily report	Draft → Submitted → Under Review → Approved / Correction Required
Material receipt	Uploaded → Verified → Posted to Stock
Order requirement	Draft → Approved → PDF Generated → Sent to Dealer → Fulfilled / Closed

6. Updated Technology Stack

Theapplication willusea practical, maintainable stack suitable for the client's initial scale. It will be deployed on the selected Hostinger KVM 1 VPS, with Amazon S3 used for file storage.

Layer	Technology	Purpose
Frontend	React.js	Admin and site-engineer interfaces.
UI styling	Tailwind CSS	Responsive, mobile-first design.
Backend	Node.js + Express.js	REST API, business logic, authentication, workflows, and integrations.
Database	PostgreSQL	Users, projects, assignments, attendance, materials, transactions, receipts, and reports.
Authentication	JWT + refresh tokens	Secure login and role-based access control.
File storage	Amazon S3	Private storage for photos, audio, receipts, and PDFs.
File upload	S3 presigned URLs	Direct uploads without passing large files through the API.
PDF generation	PDFKit	Dealer order-requirement PDFs.
Web server	Nginx	Reverse proxy, routing, compression, and HTTPS.
Process manager	PM2	Keeps the Node.js backend running.
Deployment	Hostinger KVM 1 VPS	Hosts React build, Node.js backend, PostgreSQL, and Nginx.
SSL	Let's Encrypt / Certbot	Encrypted HTTPS communication.
Backups	Automated PostgreSQL backups	Recovery of structured application data.

7. High-Level Architecture

Component	Responsibility
React frontend	Admin dashboard and site-engineer application.
Node.js API	Authentication, permissions, project logic, materials, attendance, reports, and PDF generation.
PostgreSQL	Stores structured data and file metadata.
Amazon S3	Stores progress photos, audio notes, receipts, invoices, and PDFs.
Nginx + HTTPS	Receives traffic and routes requests securely.

Important storage rule

Images, audio files, receipts, and PDFs should not be stored permanently on the VPS. PostgreSQL will store metadata such as project ID, file type, S3 key, uploader, file size, and upload date.

8. Recommended MVP Scope

Phase	Included scope
MVP 1	Login, roles, users, projects, sites, engineer-to-project assignment, and dashboards.
MVP 2	Daily progress reports, progress photos, notes, audio recordings, and review workflow.
MVP 3	Material income/outgoing, receipt uploads, material ledger, and material reports.
MVP 4	Labour register, daily attendance, labour notes, and manpower reports.
MVP 5	Admin order requirements, dealer details, and downloadable order-requirement PDFs.

Initial success criteria

- An engineer can manage multiple projects from one account.
- Daily progress can be submitted with photographs and notes.
- Labour attendance can be recorded quickly from mobile.
- Material income and outgoing remain traceable.
- Engineer-uploaded receipts are visible in the admin material section.
- Admins can generate and send dealer order-requirement PDFs.

9. Additional Inputs

In SitePulse Ops, stage-wise progress for notes is implemented by attaching a structured lifecycle state (Draft → Submitted → Under Review → Approved / Correction Required) along with metadata foreign keys (projectId, workerId, engineerId, and timestamps) to the note data structures in mockData.js.
In practice, site engineers record task observations, worker evaluations, or voice memos in the Draft stage via LabourNotesPage.jsx and DailyReportPage.jsx. Once the daily report is filed, the notes transition to Submitted and lock against field tampering. Administrators then review these entries alongside attached photo or audio evidence inside ProgressReportsPage.jsx to either send them back for revisions (Correction Required) or grant final sign-off (Approved). Finally, approved notes are permanently archived into the worker profile history, daily attendance registers, and material consumption logs for full operational and legal traceability.